CHL BACKEND API REQUIREMENTS

What The ITEIME Backend And The JRB CHL Enforcement Mobile App (MVP) Need From The Existing CHLSystem Backend

ITEM	DETAILS
Document	CHL Backend API Requirements: ITEIME Backend and Mobile MVP
Version	1.0 (Draft For The Chl Backend Team)
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Prepared For	CHL Backend Team
Requested By	Contract Developer
Related Document	CSNI-DOC-SRD-JRB CHL ENFORCEMENT (SRD v1.0), Mobile Application feature breakdown, National Enforcement Coordination & Case Management System PRD (ITEIME).
Status	Requesting Documentation and Access

1. PURPOSE

We are building our own backend, iteime_backend, for the ITEIME Enforcement Engine. It connects to the existing CHL system backend to serve the JRB CHL Enforcement mobile app. All vehicle, E-Tag, trip and agent data and every verification result come from the CHL backend; iteime_backend passes them on to the app and records what the app sends.

Scope of this request: only the connection between iteime_backend and the CHL backend for the mobile app. The ITEIME Admin Dashboard runs on iteime_backend's own data (cases, escalation, legal, finance) and does not connect to the CHL backend. The National Command Center Admin Dashboard will be covered in a separate request.

The first goal is the mobile app MVP. The app is already built and works on sample data. To make it work for real, iteime_backend must call the existing CHL backend for everything CHL already owns: agent sign-in, vehicles, RFID and QR references, E-Tags, trips and payments. This document lists exactly what we need from the CHL backend API documentation, and the access we need, to build that connection.

I will not duplicate any CHL function. Vendor, Principal Officer and agent creation, vehicle registration, RFID/E-Tag issuance, tariffs and trip rules stay in CHL (SRD Section 3.2).

1.1 HOW THE SYSTEMS CONNECT

Mobile app (MVP) ──► iteime_backend ──► Existing CHL backend (all data and results)
ITEIME Admin Dashboard ──► iteime_backend only (own data, no CHL connection)
National Command Center Admin Dashboard ──► separate request (not covered here)

The mobile app never calls CHL directly. iteime_backend calls CHL server-to-server, so we need server-to-server access as well as the agent sign-in flow.

1.2 WHAT WE NEED, IN SHORT

1. The API documentation (OpenAPI/Swagger file or Postman collection) for the endpoints in section 3.
2. A test (staging) environment: base URL, server-to-server credentials and test agent accounts.
3. Test data: sample vehicles, tags and QR codes covering every result in section 4.
4. Answers to the open questions in section 8.

2. GENERAL API INFORMATION WE NEED

#	WHAT WE NEED	WHY
G1	Base URLs For Test (Staging) And Production	To Configure Iteime_backend Per Environment
G2	Api Specification: Openapi 3 (Swagger) File, Or A Postman Collection With Example Requests And Responses	To Build And Test The Chl Adapter Accurately
G3	Server-To-Server Authentication: Api Key, Oauth2 Client Credentials, Or Other; How Keys Are Issued And Rotated	Iteime_backend Calls Chl On Behalf Of The App
G4	Ip Allow-Listing Or Vpn Requirements, If Any	To Reach Chl From Our Servers
G5	How Agent Sign-In Works: Which Endpoint Verifies An Agent's Email And Password, And Whether Chl Issues The Agent Tokens Or Iteime_backend Should	Agents Sign In With Their Existing Chl Credentials (Srd §5)
G6	Request And Response Format (Json), Character Encoding, Field Naming (Camelcase Or Snake_case)	Consistent Parsing
G7	Date/Time Format And Time Zone (E.G. Iso 8601 With Offset, Or Utc)	Trip Times, Countdowns And Audit Records

G8	Standard Error Format And The Full List Of Error Codes; Http Status Usage (400, 401, 403, 404, 409, 422, 429, 5xx)	To Tell "Not Found" From "Service Failure" (SRD Section 15 & 26)
G9	Rate Limits And Expected Response Times	To Size Timeouts And Retries
G10	Api Versioning And Change Notice Policy	To Avoid Breaking The App
G11	Pagination And Filtering Conventions (For List Endpoints)	Lists Of Trips, Payments
G12	Idempotency Support (E.G. An Idempotency-Key Header) For Write Calls	Offline Sync Must Not Create Duplicates (SRD Section 33.2)
G13	Webhooks Or Events Chl Can Send To Us (E.G. Payment Confirmed, Trip Activated, Account Deactivated)	Live Payment Status And Immediate Access Denial
G14	A Technical Contact On The Chl Team	For Questions During Integration

3. CAPABILITIES NEEDED FOR THE MOBILE MVP

For each capability: what the mobile app does, what we need CHL to provide, and the fields we need in the response. Endpoint names are ours, for reference only; please tell us the real CHL endpoints.

3.1 AGENT AUTHENTICATION AND SESSION (SRD)

Mobile app: sign-in with email and password, Forgot Password, 24-hour session, immediate denial for a deactivated account, sign-out.

#	CAPABILITY	INPUT	WHAT WE NEED BACK
A1	Sign In An Agent	Email, Password	Success Or Failure; Tokens (Access, Refresh, Expiry) If Chl Issues Them; The Agent Profile (A5) And The Basic Role Validation
A2	Refresh A Session	Refresh Token	New Access Token And Expiry
A3	Sign Out / Revoke	Token	Confirmation
A4	Forgot Password	Email	Confirmation That The Reset Email Was Sent (The Reset Itself Stays In Chl)

A5	Agent Profile	Agent / Token	Agent Id (E.G. Jrb-Ag-0142), Full Name, Email, Role, Organisation/Vendor, Region, Assigned Enforcement Scope (Location, Zone, Ward, LGA, State, Region, Subregion, National), Account Status
A6	Account Status Check	Agent Id Or Token	Active / Deactivated, So Access Is Denied At Once When An Account Is Deactivated Centrally

Also tell us:

- The exact role names (e.g. ENFORCEMENT_AGENT, PRINCIPAL_OFFICER, SUPERVISOR, SYSTEM_ADMINISTRATOR). Only enforcement agents may use the mobile app.
- The error codes for: wrong password, unknown email, deactivated account, locked account, too many attempts.
- Whether sessions end after 24 hours on the CHL side, or iteime_backend should enforce it.

3.2 VEHICLE VERIFICATION (SRD SECTION 8 & 13)

Mobile app: four methods (RFID haulage tag, QR code, manual plate, OCR plate) that all resolve to the same verification. The app sends the agent's GPS with every verification.

#	CAPABILITY	INPUT	WHAT WE NEED BACK
V1	Verify By Plate Number	Normalised Plate (Spaces, Hyphens Removed, Upper Case, E.G. Abc123aa)	The Verification Response (V6)
V2	Verify By Qr Reference	Reference Read From The E-Tag Qr Code	The Verification Response (V6)
V3	Verify By Rfid Tag	Tag Id And/Or Reference Number Read From The Tag (Please Tell Us Which One Chl Expects; Srd Section (9.4)	The Verification Response (V6)
V4	Resolve Several Rfid Tags At Once	A List Of Tag References	For Each: The Vehicle Plate It Belongs To, Or "Unknown Tag" (Srd Section 9.6 Multiple Tags)
V5	Recognise An Unknown Tag	Tag Reference	A Clear "Not Part Of The Jrb Chl Architecture" Answer, Different

			From "Vehicle Not Found" (SRD Section 9.7, 13.2)
V6	Verification Response	Null	Vehicle: Id, Plate, Vehicle Type (E.G. "12/14 Tyres And Its Equivalent"). E-Tag Status: Valid / Invalid / Missing / Mismatch (And The Plate The Tag Belongs To, When Different). Trip Status: Active / No Active Trip / No Trip History / Expired Or Previous. The Current Trip (3.3) When Active; The Most Recent Trip (3.3) When Not. The Destination-Check Details (3.4) For An Active Trip. A Chl Verification Reference, If Chl Keeps One.

3.3 CHL TRIP DETAILS (SRD SECTION 16 & 17)

The app shows these fields for the active trip and for the last trip. Please confirm each field name and format.

FIELD	EXAMPLE	NOTES
Transaction Reference	564031956608	
Trip Start Date/Time	October 5, 2026 4:28 Pm	With Time Zone
Transaction (Payment) Date/Time	October 5, 2026 4:22 Pm	
Vehicle License (Plate) Number	Jmt150xb	
Vehicle Type	12/14 Tyres And Its Equivalent	
Transaction Type (Trip Type)	Complete Leg	List Of Possible Values
Transaction Status	Successful	List Of Possible Values
Originating State / Lga	Ogun / Abeokuta South	
Destination State / Lga	Lagos / Ikeja	
Trip Status	Active / Completed / Expired	As Chl Reports It; List Of Values

Trip Will Close On (Valid Until)	October 19, 2026 11:59 Pm	Drives The Validity Countdown
Completion Recorded At	October 5, 2026 9:45 Pm	For Completed Trips
Payment Reference	Pay-2026-08142	
Terminal Id	2070mjhu	From The Receipt
Contact Number	8139751699	From The Receipt
Channel	Fortis	From The Receipt
Channel Type	Pos	From The Receipt

#	CAPABILITY	INPUT	WHAT WE NEED BACK
T1	Get The Current (Active) Trip	Vehicle Id Or Plate	The Trip, Or "No Active Trip"
T2	Get The Most Recent Trip	Vehicle Id Or Plate	Only The Most Recent Trip, Or "No Trip History" (Srd §17)
T3	Refresh Trip Status	Vehicle Id Or Transaction Reference	The Latest Trip And Status (The App's Refresh Trip Status Button)

3.4 DESTINATION CHECK (SRD SECTION 14.2 & 25)

#	CAPABILITY	INPUT	WHAT WE NEED BACK
D1	Destination-Check Details For An Active Trip	Transaction Reference	Enforcement Point Name, Destination Lga, Completion Decision (E.G. "Awaiting Central Validation"), The Centrally Defined Delay (E.G. 900 Seconds)
D2	Record The Agent's Location For Destination Detection	Transaction Reference, Latitude, Longitude, Accuracy, Time, Agent Id	The Location As Chl Received It, And Any Updated Completion Decision
D3	Enforcement Points	Null	The List Of Registered Enforcement Points With Their Locations / Geofences, To Tag Every Verification With Its Enforcement Point

3.5 LOADED / EMPTY CONFIRMATION (SRD SECTION 2 & 13.4)

#	CAPABILITY	INPUT	WHAT WE NEED BACK
L1	Record Whether A Vehicle With No Active Trip Is Loaded Or Empty	Vehicle Id Or Verification Reference, Loaded Or Empty, Agent Id, Gps, Time	Confirmation. Please Say Whether Chl Stores This, Or Iteime_backend Should Store It And Report It.

3.6 E-TAG RECEIPT (QR ON THE E-TAG)

#	CAPABILITY	INPUT	WHAT WE NEED BACK
E1	Get The E-Tag Receipt	E-Tag Reference (From Its Qr Code)	All Receipt Fields Chl Holds For That E-Tag (Please List Them), Or "Not Found"
E2	E-Tag Status	E-Tag Reference	Active / Damaged / Not Picked Up / Replaced, And The Vehicle It Is Installed On (To Support Tag-Vehicle Mismatch)

3.7 REFERENCE DATA

#	DATA	USE
R1	States And LAGs (With Codes)	Display And Filtering
R2	Vehicle Types	Display
R3	Transaction Types, Transaction Statuses And Trip Statuses (All Possible Values)	Correct Labels In The App
R4	Enforcement Zones, Wards, Regions, Subregions, And How They Nest	Agent Scope And Dashboards

3.8 HELPFUL FOR THE MVP (PLEASE TELL US IF AVAILABLE)

#	CAPABILITY	WHY
O1	Payment Status For A Transaction Or Payment Reference	Live Payment Confirmation (After The Mvp, For Demand Notices)
O2	Notifications Chl Sends To Agents (E.G. Repeat-Vehicle Alerts)	The App's Notifications Inbox

O3	Data For An Offline Cache (Recent Vehicles And Active Trips By State)	Offline Verification (SRD Section 26)

4. RESULT STATES CHL MUST TELL APART

The SRD requires that an API failure is never shown as "No Active Trip" (SRD §15, §26.2, acceptance criterion 12). Please give each of these a distinct response (status code and/or result code):

RESULT	MEANING	MOBILE APP SCREEN
Active Trip	Vehicle Identified, Active Chl Trip	Active Chl Trip
No Active Trip	Vehicle Identified, No Current Trip (Most Recent Trip Returned)	No Active Trip Found → View Last Chl Trip
No Trip History	Vehicle Exists, Has Never Made A Chl Trip	Last Chl Trip: No Trip History
Vehicle Not Found	Plate / Qr / Tag Cannot Be Resolved To A Vehicle	Vehicle Not Found
Invalid E-Tag	Tag Is Not Part Of The Jrb Chl Architecture	Tag Not Part Of Jrb Chl Architecture
Tag-Vehicle Mismatch	Tag Belongs To A Different Vehicle	(Coming)
E-Tag Missing	No Tag On The Vehicle (Agent Confirms)	(Coming)
Expired/Previous Trip	Previous Trip No Longer Current	Last Chl Trip
Service Unavailable	Chl Could Not Complete The Check (Timeout, Error)	Verification Unavailable

5. DATA FORMATS TO CONFIRM

ITEM	WHAT WE ASSUME TODAY	PLEASE CONFIRM
Plate Number	3–10 Letters And Digits; Spaces, Hyphens And Case Ignored	The Official Rules And Examples

Rfid Tag Reference	24 Hexadecimal Characters Starting "E2" (Epc-96)	What The Reader Returns And What Chl Expects (Tag Id Vs Reference Number)
Qr Code On The E-Tag	Contains The Vehicle's Reference Number	The Exact Content And Format
Dates And Times	Iso 8601 With Time Zone	Format And Time Zone
Coordinates	Decimal Degrees (Wgs84)	Precision And Accuracy Fields
IDs	Agent Id Like Jrb-Ag-0142; Transaction Reference Numeric	Formats And Uniqueness

6. AUDIT FIELDS SENT WITH EVERY VERIFICATION (SRD SECTION 28)

Iteime_backend Records Every Verification Attempt. Please Tell Us Which Of These CHL Also Wants To Receive (If Any), And In What Format:

- Agent Id, Agent Name, Vendor/Organisation
- Date And Time; Verification Method (RFID, QR, Plate, OCR)
- Processed Identifier (Never Shown To Agents)
- Vehicle Id And Plate Where Resolved; Verification Result
- Trip Reference; Violation Reference Where Applicable
- Gps Location And Enforcement Point
- Device And App Metadata; Action Taken; Synchronisation Status (Live Or Offline)

7. TEST ACCESS AND TEST DATA WE NEED

7.1 TEST AGENT ACCOUNTS

ACCOUNT	PURPOSE
Active Enforcement Agent	Normal Sign-In
Deactivated Agent	Access Denied
Account With A Non-Agent Role	Agent Access Not Authorised
Agent With A Forgot-Password Test Email	Password Recovery

7.2 TEST VEHICLES, TAGS AND QR CODES

SCENARIO	NEEDED AS
Vehicle With An Active Trip And A Valid E-Tag	Plate, Tag Reference, Qr
Vehicle With No Active Trip And A Previous Trip	Plate, Tag Reference, Qr
Vehicle With No Trip History	Plate
Unknown Plate (Vehicle Not Found)	Plate
Tag Not Part Of The Jrb Chl Architecture	Tag Reference
Tag That Belongs To Another Vehicle (Mismatch)	Tag Reference + Plate
Two Tags Of Different Vehicles Read Together	Two Tag References
Expired Trip	Plate
A Way To Simulate Chl Being Unavailable	Test Switch Or Endpoint

8. OPEN QUESTIONS FOR THE CHL TEAM

1. Does Chl Issue The Agent's Session Tokens, Or Does Iteime_backend Verify Credentials With Chl And Issue Its Own Tokens?

2. Which Identifier Does Chl Expect From An Rfid Read: The Tag Id, The Reference Number, Or Both?

3. Is There A Batch Endpoint To Resolve Several Tag References To Plates In One Call?

4. Does Chl Already Return A Single Combined Verification (Vehicle + E-Tag + Trip), Or Must We Call Separate Endpoints And Combine Them?

5. How Does Chl Report "No Trip History" Versus "No Active Trip"?

6. Does Chl Store Loaded/Empty Confirmations And Destination-Check Locations, Or Should Iteime_backend Store Them?

7. Which E-Tag Receipt Fields Does Chl Hold?

8. Can Chl Notify Iteime_backend (Webhook/Event) When An Account Is Deactivated, A Payment Is Confirmed Or A Trip Is Activated?

9. What Are The Rate Limits And The Expected Response Time For Verification?

10. Is There A Sandbox We Can Use Before Production Access, And How Long Does Access Take To Set Up?

9. SUMMARY CHECKLIST

#	ITEM	PRIORITY FOR THE MVP
1	Api Specification (Openapi / Postman) And Staging Base Url	Must Have
2	Server-To-Server Credentials	Must Have
3	Agent Sign-In, Refresh, Sign-Out, Forgot Password, Profile, Account Status (3.1)	Must Have
4	Verify By Plate, Qr And Rfid Tag; Multiple-Tag Resolution; Unknown Tag (3.2)	Must Have
5	Current Trip, Most Recent Trip, Refresh (3.3)	Must Have
6	Distinct Result States And Error Codes (Section 4, G8)	Must Have
7	Test Accounts And Test Vehicles (Section 7)	Must Have
8	Destination Check And Enforcement Points (3.4)	Should Have
9	Loaded/Empty Confirmation (3.5)	Should Have
10	E-Tag Receipt And E-Tag Status (3.6)	Should Have
11	Reference Data (3.7)	Should Have
12	Payment Status, Notifications, Offline Cache Data (3.8)	Nice To Have (After Mvp)

Thank you. Please send the documentation and access details. I will build against a mock of CHL until they arrive.